

MBTiles to TPKX Converter

Official User Manual

Purpose

Convert an existing raster MBTiles file directly to a native Esri TPKX for offline viewing in ArcGIS Earth. Source PNG/JPEG map tiles are copied unchanged; the converter does not download imagery or rebuild the zoom pyramid.

Requirements

Windows, Python 3, Pillow, and an existing compatible raster MBTiles file.

```
py -3 --version
```

```
py -3 -m pip install Pillow
```

Keep these together

mb2tpkx.py

mb2tpkx.bat

mb2tpkx_User_Manual.pdf

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Run it

1. Put the Python script and BAT launcher in the same folder.
2. Drag your .mbtiles file onto mb2tpkx.bat, or double-click the BAT and paste the full path.
3. Wait for SUCCESS. The TPKX is created beside the MBTiles with the same base name.
4. Open the finished TPKX in ArcGIS Earth and verify the map.

Command line

```
py -3 mb2tpkx.py "input.mbtiles" "output.tpkx"
```

If output is omitted, the input name is reused with .tpkx. Existing output is not overwritten.

Supported input

Standard Web Mercator raster MBTiles, 256 x 256 PNG/JPEG tiles, TMS tile rows, zoom levels 0-23. Vector MBTiles and arbitrary projections are outside the verified scope.

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What it does

Reads MBTiles, converts TMS rows to Esri XYZ rows, groups tiles into 128 x 128 Compact Cache V2 bundles, writes TPKX metadata, and copies source tile bytes unchanged. Only a separate presentation thumbnail is generated.

Common errors

Pillow is required: run the install command on page 1.

MBTiles file not found: verify the path or drag the file onto the BAT.

Output already exists: rename, move, or remove the previous TPKX.

Unsupported tile: input must use 256 x 256 PNG/JPEG raster tiles.

Zoom exceeds verified scheme: current implementation supports 0-23.

Large files

Allow enough time and free disk space. If a conversion is interrupted, discard the incomplete output before restarting.

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QGIS workflow

1. Create the raster MBTiles in QGIS using your intended source, extent, zoom range, and image format.
2. For large systematic coverage, use Master 4 Grid Maker for exact repeatable extents and filenames.
3. Convert each MBTiles with mb2tpkx.bat.
4. Verify the resulting TPKX in ArcGIS Earth before field use.

Quick reference

Install: `py -3 -m pip install Pillow`

Run: drag `input.mbtiles` onto `mb2tpkx.bat`

Output: `input.tpkx` beside `input.mbtiles`

Input MBTiles remains unchanged. Existing TPKX is protected from overwrite.

Source rights

This utility does not grant rights to third-party imagery. Use only imagery you are authorized to acquire, retain, convert, display, or redistribute.